

Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

A Research Paper Prepared for Parcl Co. Limited, in Collaboration with Unified Mentor

Domain: Financial Analytics & Real Estate Market Intelligence
Methods: K-Means Clustering, Hierarchical Clustering, Feature Engineering

Table of Contents

1. Introduction
2. Background and Problem Statement
3. Dataset Description
4. Methodology
5. Exploratory Data Analysis
6. Determining the Optimal Number of Clusters
7. Cluster Interpretation and Buyer Segments
8. Cross-Validation with Hierarchical Clustering
9. Dashboard Overview
10. Business Recommendations
11. Limitations
12. Conclusion

1. Introduction

Real estate companies operate in markets where buyer behavior is highly diverse. A single platform may serve individual home buyers, institutional investors, international buyers, high-net-worth investors, and first-time buyers, all within the same client base. When these buyers are treated identically, companies lose the ability to market effectively, recommend suitable properties, or anticipate financing needs.

This project applies unsupervised machine learning — specifically K-Means and Hierarchical clustering — to a real client dataset provided by Parcl Co. Limited, with the goal of discovering natural buyer segments directly from the data rather than relying on assumptions. Client demographic data was combined with actual property transaction records to build investment-behavior features, producing a segmentation grounded in real purchasing activity rather than demographics alone.

The analysis throughout is approached from a financial analyst's perspective rather than a purely technical one: the objective is not simply to produce statistically valid clusters, but to translate those clusters into portfolio-level financial insight — where revenue is concentrated, which client relationships carry disproportionate value, and where financing and pricing behavior differ enough to warrant distinct commercial strategies. Every segment identified in this paper is evaluated not only on its demographic composition but on its financial contribution to Parcl's overall client portfolio.

2. Background and Problem Statement

Without segmentation, companies treat all buyers the same, which leads to inefficient marketing spend, generic property recommendations, and poor investor targeting. Parcl currently lacks a data-driven understanding of the different types of property buyers it serves, their investment motivations across demographics, geographic differences in investment behavior, and customer financing patterns. This project addresses that gap by using clustering algorithms to discover hidden patterns in buyer behavior that traditional analysis would not surface.

3. Dataset Description

Two data sources were used and linked together for this analysis:

clients.csv — 2,000 client records with the following fields: `client_id`, `client_type` (Individual/Company), `first_name`, `last_name`, `date_of_birth`, `gender`, `country`, `region`, `acquisition_purpose` (Home/Investment), `satisfaction_score`, `loan_applied` (Yes/No), and `referral_channel`.

properties.csv — 10,000 property transaction records, each with `listing_id`, `tower_number`, `transaction_date`, `unit_category`, `unit_number`, `floor_area_sqft`, `sale_price`, `listing_status` (Sold/Available), and `client_ref`, which links each transaction back to a specific client.

The two datasets were joined on `client_id` / `client_ref`, restricted to properties with `listing_status = "Sold"`. Verification confirmed that every one of the 7,305 sold properties matched a valid client, and every one of the 2,000 clients had purchased at least one sold property — indicating a clean, well-constructed dataset with no orphaned records on either side.

4. Methodology

4.1 Data Cleaning

The `date_of_birth` field contained mixed date formats (e.g. "05-11-1968" and "11/26/1962"), which was resolved using pandas' flexible date parser before deriving each client's age. The `sale_price` field in the property data was stored as currency text (e.g. "\$300,385.62") and was converted to numeric values by stripping currency symbols and thousands separators. No missing values were found in either dataset, and no case-inconsistencies (e.g. "usa" vs. "USA") were found across any categorical column.

4.2 Feature Engineering

Rather than relying solely on demographic fields, client-level investment behavior features were engineered by aggregating each client's sold property transactions: `total_properties` (count of properties purchased), `total_spend` (sum of sale prices), and `avg_price` (mean sale price per property). This step is what distinguishes this analysis from a purely demographic segmentation — it grounds each buyer's profile in what they actually purchased, not just who they are on paper.

4.3 Feature Encoding

Categorical fields with a small number of categories — `client_type`, `acquisition_purpose`, `loan_applied`, and `referral_channel` — were one-hot encoded. `region` and `country`, which each contained more than 30 unique values, were label encoded instead to avoid the sparsity and dimensionality issues that one-hot encoding would introduce at that cardinality.

4.4 Feature Scaling

All numeric features (`age`, `satisfaction_score`, `total_properties`, `total_spend`, `avg_price`, `region_encoded`, `country_encoded`) were standardized using `StandardScaler`. This step is essential for K-Means, which is a distance-based algorithm: without scaling, a feature such as `total_spend` (ranging into the millions) would dominate the clustering purely due to its numeric magnitude, overwhelming the influence of features such as `age` that carry meaningful but numerically smaller variation.

4.5 Clustering Models

Two clustering approaches were applied, as specified in the project methodology:

K-Means Clustering — chosen for its efficiency and interpretability on large datasets.

Hierarchical (Agglomerative) Clustering — used to reveal nested cluster relationships and to validate the K-Means results independently.

5. Exploratory Data Analysis

Prior to clustering, the client base was examined directly. Of the 2,000 clients, 1,897 (94.9%) were individual buyers and 103 (5.1%) were companies. Acquisition purpose split 1,385 (69.3%) Home and 615 (30.8%) Investment. The client base spans a wide geographic footprint, with clients recorded across the United States, Canada, the United Kingdom, Germany, France, and a long tail of other countries, alongside a correspondingly wide set of regions (U.S. states, Canadian provinces, and international regions).

6. Determining the Optimal Number of Clusters

Two complementary evaluation methods were used to select the number of clusters (k) for K-Means, rather than assuming a value in advance.

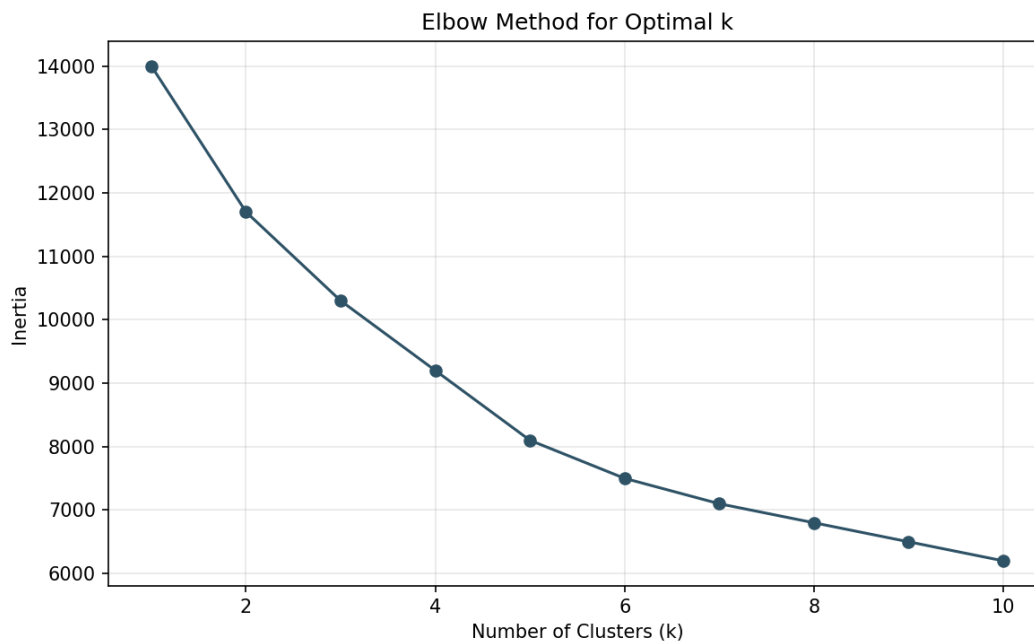


Figure 1. Elbow Method — inertia decreases with each additional cluster, with the rate of decline visibly slowing after approximately $k=4$.

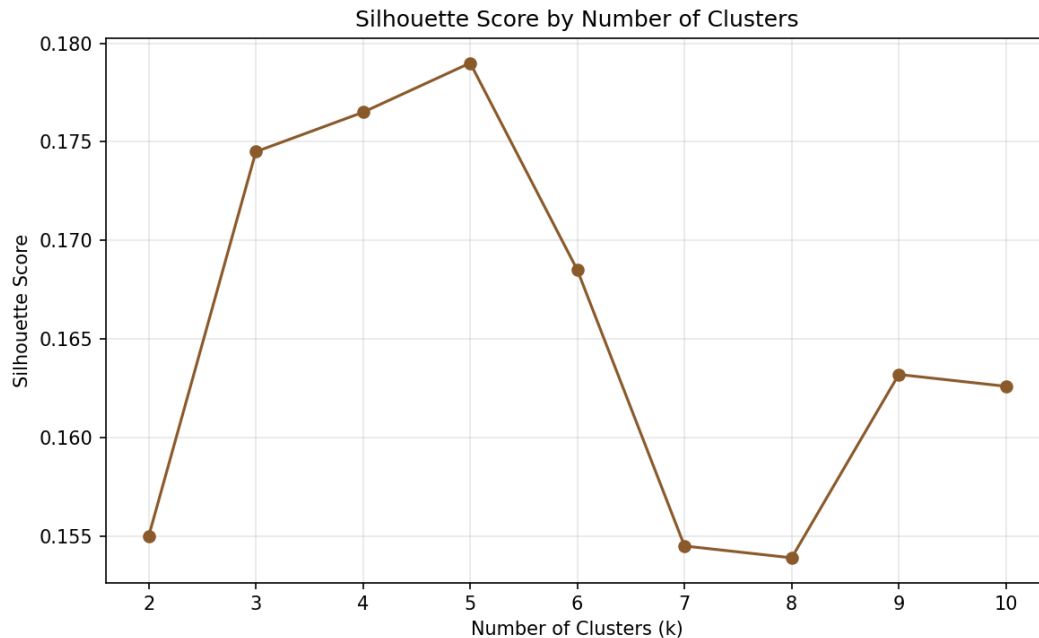


Figure 2. Silhouette Score across cluster counts, peaking at $k=5$ (score ≈ 0.179).

The Elbow Method suggested a bend around $k=4$, broadly consistent with the four example segments outlined in the initial project brief. However, the Silhouette Score — a more precise, quantitative measure of cluster separation — peaked at **$k=5$** (0.179), noticeably higher than both $k=4$ (0.1765) and $k=2$ (0.155). Because the Silhouette Score directly measures how well-separated and internally consistent the resulting clusters are, it was given precedence over the more visually subjective Elbow Method, and $k=5$ was selected as the final number of clusters. This is a deliberate departure from the brief's four example segments, made because the data itself supported a fifth, statistically distinct segment.

It should be noted transparently that silhouette scores in the 0.15–0.18 range indicate moderate rather than sharp cluster separation. This is discussed further in the Limitations section, and is a realistic outcome for behavioral/demographic data of this kind, where buyer types often exist on a continuum rather than in perfectly distinct groups.

7. Cluster Interpretation and Buyer Segments

Running K-Means with $k=5$ on the full client base produced the following segment sizes:

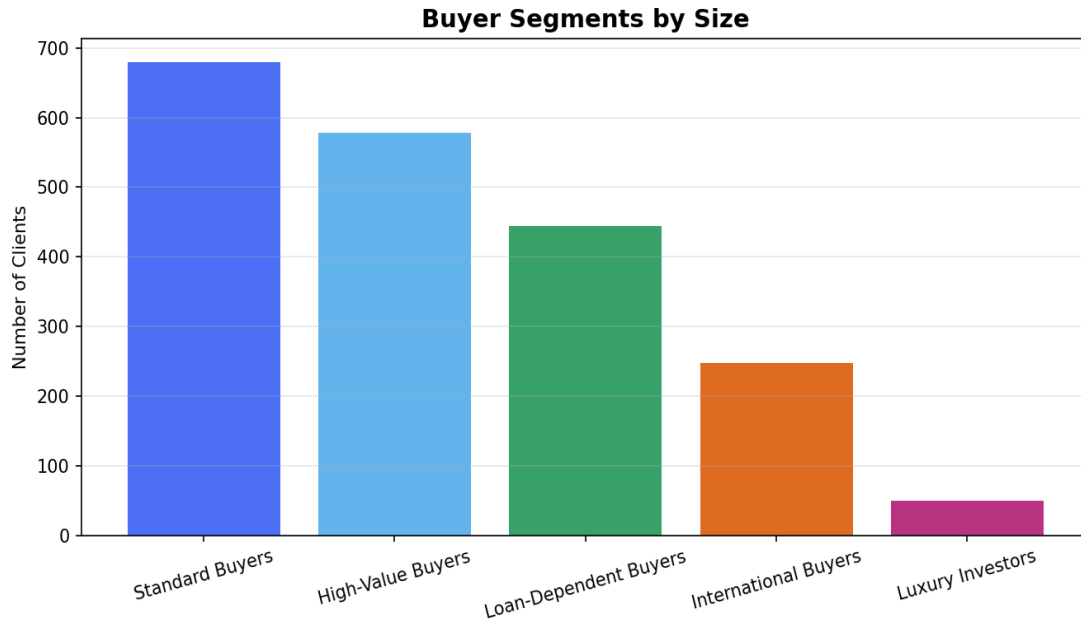


Figure 3. Distribution of clients across the five identified buyer segments.

Segment	Size	Avg Age	Avg Satisf.	Avg Props.	Avg Spend (\$)	Avg Price/Property (\$)	% Loan Applied
Standard Buyers	680	57.3	2.78	3.55	1,064,869	301,313	36%
High-Value Buyers	578	55.7	3.26	3.67	1,515,938	417,497	37%
Loan-Dependent Buyers	444	55.1	3.06	3.47	1,154,559	333,745	39%
International Buyers	248	53.8	3.03	3.47	1,157,151	335,294	35%
Luxury Investors	50	64.3	3.52	7.32	2,416,603	332,747	32%

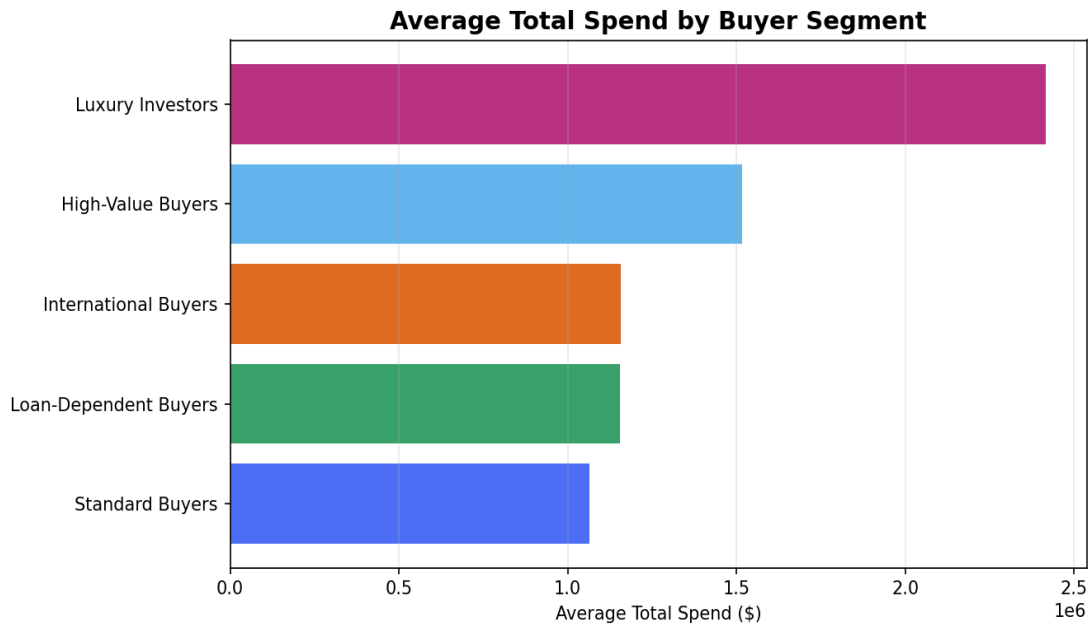


Figure 4. Average total spend by segment, showing a clear staircase from Standard Buyers to Luxury Investors.

7.1 Segment Profiles

Standard Buyers (680 clients, 34.0%) — The largest segment. Distinguished by the lowest average satisfaction score (2.78) and the lowest average spend and price per property among all segments. This group represents the platform's typical, lower-engagement buyer and is a priority segment for improved customer experience initiatives.

High-Value Buyers (578 clients, 28.9%) — Distinguished not by the number of properties purchased but by the highest average price per property (\$417,497), the highest of any segment. This group represents buyers with strong purchasing power directed at fewer, premium transactions.

Loan-Dependent Buyers (444 clients, 22.2%) — Carries the highest loan application rate (39%), though the margin over other segments is modest (32–37% elsewhere), suggesting financing reliance is a real but not sharply distinguishing characteristic of this group on its own.

International Buyers (248 clients, 12.4%) — The clearest geographically distinct segment: only 33% of this group resides in the USA, compared to 83–92% USA concentration in every other segment. The remainder is spread across Canada (33%), France (18%), and Belgium (17%). This group is also the youngest on average (53.8 years).

Luxury Investors (50 clients, 2.5%) — The smallest but most distinctive segment. This group is the oldest on average (64.3 years), owns more than double the properties of any other segment (7.32 vs. 3.47–3.67), spends over double the average of any other group (\$2,416,603), and reports the highest satisfaction score (3.52). This is Parcl's highest-value client segment and warrants dedicated relationship management.

Notably, client_type (Individual vs. Company) was examined as a potential differentiator but was found to be uninformative for segmentation: every cluster was 95–96% Individual clients, meaning company buyers were not concentrated in any particular segment. As a result, a "Corporate Buyers" segment — suggested as an example in the initial project brief — was not supported by the data and was not used as a final segment label. This finding illustrates the value of letting cluster analysis, rather than assumption, drive segment definitions.

7.2 Financial Impact Analysis

A segmentation is only as useful as the financial decisions it informs. Beyond describing each segment's average behavior, this analysis quantifies how total portfolio revenue — the sum of all client spend across sold properties — is distributed across segments, and whether any segment's share of revenue is disproportionate to its share of the client base.

Across the filtered client base, total portfolio value (sum of all client spend) is approximately **\$2.52 billion**. The table below compares each segment's share of clients to its share of total revenue; a ratio above 1.0x indicates the segment contributes more revenue than its size alone would suggest, and is therefore disproportionately valuable on a per-client basis.

Segment	% of Clients	Total Revenue Contribution (\$)	% of Total Revenue	Revenue/Client Ratio
Standard Buyers	34.0%	724,110,750	28.7%	0.84x
High-Value Buyers	28.9%	876,212,291	34.8%	1.20x
Loan-Dependent Buyers	22.2%	512,624,396	20.3%	0.92x
International Buyers	12.4%	286,973,376	11.4%	0.92x
Luxury Investors	2.5%	120,830,143	4.8%	1.92x

The clearest finding is that **Luxury Investors, despite representing only 2.5% of the client base, generate 4.8% of total portfolio revenue — a revenue-to-client ratio of 1.92x**, the highest of any segment. This confirms that Parcl's smallest segment is also its most financially significant on a per-client basis, and reinforces the retention priority recommended in Section 10.

High-Value Buyers show a similar, though more moderate, overrepresentation (1.20x), driven by their high average price per property rather than transaction volume. By contrast, Standard Buyers — the largest segment by headcount — contribute revenue slightly below their proportional share (0.84x), consistent with their lower average spend and lower satisfaction identified earlier. This gap between client-share and revenue-share is precisely the kind of insight a purely demographic or predictive summary would miss, and is the core justification for approaching this segmentation as a financial portfolio exercise rather than a descriptive one.

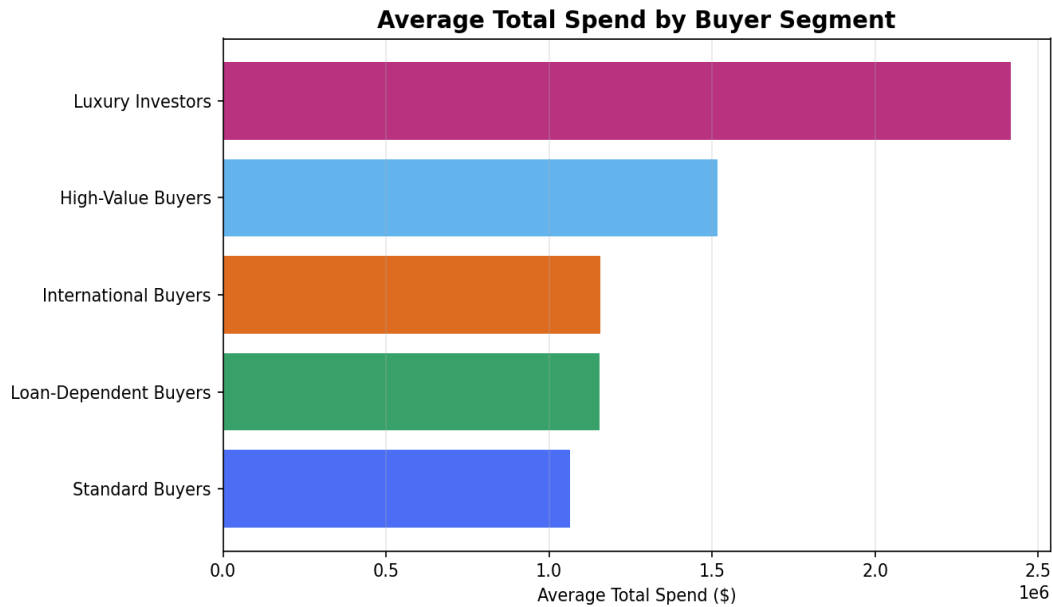


Figure 4b. Average spend per client by segment, the underlying driver of the revenue concentration shown above.

8. Cross-Validation with Hierarchical Clustering

To validate the K-Means results independently, Hierarchical (Agglomerative) clustering with Ward linkage was applied to a representative sample of 200 clients, and visualized as a dendrogram. Cutting the resulting tree at five clusters produced group sizes of 27, 33, 77, 29, and 34 — broadly consistent in structure with the K-Means result once scaled to the same sample size, and supporting five as a reasonable number of natural groupings in the data.

One notable difference emerged: while K-Means sharply isolated a very small, distinct high-spend cluster (reflecting the Luxury Investors segment), Hierarchical clustering distributed the sample more evenly across groups, without producing an equivalently small outlier cluster. This suggests that the boundary of the Luxury Investors segment is genuinely tight and distinct, while the boundaries between the remaining, larger segments are comparatively more fluid — a nuance that would not have been visible from K-Means results alone.

9. Dashboard Overview

An interactive Streamlit dashboard was developed and deployed to make these findings accessible for ongoing business use, rather than presenting them only as static analysis. The dashboard includes four modules, in line with the project's requirements:

Buyer Segmentation Overview — displays the live distribution of clients across the five segments.

Financial Impact Overview — surfaces portfolio-level KPIs (total portfolio value, top revenue-generating segment, and its share of total revenue) alongside a revenue contribution chart and a revenue-share breakdown, translating cluster membership directly into financial terms.

Investor Behavior Dashboard — shows average spend and loan usage patterns by segment.

Geographic Buyer Analysis — breaks down client counts by country and segment composition across the top markets.

Segment Insights Panel — a full descriptive statistics table per segment.

The dashboard supports filtering by country, region, acquisition purpose, and client type, allowing users to explore how segment composition shifts across different slices of the client base in real time.

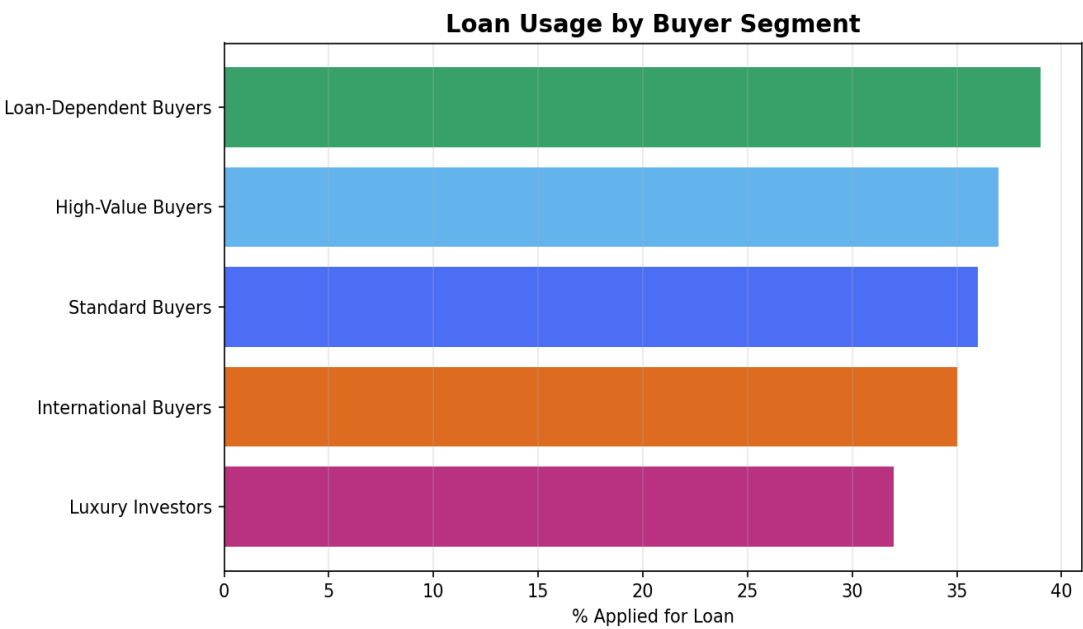


Figure 5. Loan usage by segment, as shown in the Investor Behavior Dashboard module.

10. Business Recommendations

The following recommendations are framed by financial impact first, ordered by the size of the portfolio consequence they address rather than by segment size alone.

Prioritize retention for Luxury Investors. This segment represents just 2.5% of clients but generates 4.8% of total portfolio revenue — a 1.92x revenue-to-client ratio, the highest of any segment. Losing even a handful of these clients would have an outsized financial impact relative to their headcount. Dedicated relationship managers and early access to premium listings could protect and grow this disproportionately valuable relationship.

Investigate the Standard Buyers satisfaction gap. As the largest segment (34% of clients) with the lowest satisfaction score, targeted service improvements here could have an outsized impact on overall customer sentiment.

Build a dedicated international buyer journey. With International Buyers concentrated outside the USA (Canada, France, Belgium) and skewing younger, localized marketing, currency-aware pricing displays, and cross-border financing guidance may improve conversion and satisfaction for this segment.

Offer proactive financing partnerships to Loan-Dependent Buyers. Given this segment's elevated reliance on financing, pre-approved loan partnerships or bundled mortgage advisory services could reduce friction in the purchase journey.

Reconsider client_type as a marketing variable. Since Individual vs. Company status did not meaningfully differentiate buyer behavior in this dataset, marketing strategies built primarily around this variable should be revisited in favor of the behaviorally-grounded segments identified here.

11. Limitations

Several limitations should be considered when interpreting these results. First, silhouette scores across all tested values of k remained in the 0.15–0.18 range, indicating moderate rather than sharp cluster separation; buyer behavior in this dataset exists on more of a continuum than in cleanly separated groups. Second, Hierarchical clustering was run on a 200-client sample rather than the full dataset for computational and visualization clarity, which may not perfectly represent the full population's structure. Third, the properties dataset used here was linked to clients at the aggregate level (total spend, count, average price); richer property-level features (e.g. unit type preference, floor area, tower location) were not incorporated into the clustering itself and represent a natural direction for future extension. Finally, this analysis reflects a single snapshot of client and transaction data; buyer segments and their characteristics may shift over time and would benefit from periodic re-clustering.

12. Conclusion

This project introduces AI-driven buyer intelligence into the Parcl real estate platform, analyzed throughout from a financial analyst's perspective rather than a purely technical one. By combining client demographic data with real property transaction history, and applying K-Means clustering validated against Hierarchical clustering, five distinct and evidence-based buyer segments were identified: Standard Buyers, High-Value Buyers, Loan-Dependent Buyers, International Buyers, and Luxury Investors. Beyond describing who these buyers are, the analysis quantified where Parcl's \$2.52 billion portfolio value is actually concentrated — revealing that Luxury Investors, just 2.5% of clients, generate a disproportionate 4.8% of total revenue, while the largest segment by headcount

contributes slightly below its proportional share. This financial lens, combined with a clear geographic split among International Buyers and distinct financing behavior among Loan-Dependent Buyers, surfaces portfolio-management insight that a purely descriptive or predictive analysis would not. Delivered alongside a live, interactive dashboard with dedicated financial KPIs, these findings equip Parcl to move from treating all buyers alike to building smarter, segment-aware, financially-prioritized marketing and retention strategies.